# One-Way ANOVA

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## Introduction

One-way ANOVA tests whether the means of three or more independent groups are equal. It generalises the two-sample t-test to multiple groups while controlling overall Type I error. The omnibus F-test tells us whether *any* means differ; post-hoc tests (Tukey HSD, Dunnett) identify which pairs.

## Prerequisites

Two-sample t-test, F distribution.

## Theory

For $k$ groups with $n_i$ observations each, total $n = \sum n_i$, define:

- **Sum of squares between** (SSB): $\sum_i n_i(\bar{X}_i - \bar{X})^2$ with $k-1$ df.
- **Sum of squares within** (SSW): $\sum_i \sum_j (X_{ij} - \bar{X}_i)^2$ with $n-k$ df.
- **Mean squares**: MSB = SSB / (k - 1), MSW = SSW / (n - k).

Under $H_0 : \mu_1 = \ldots = \mu_k$,

$$F = \frac{\text{MSB}}{\text{MSW}} \sim F_{k-1,n-k}.$$

Large F implies between-group variation exceeds within-group variation, rejecting equality of means.

Effect size: $\eta^2 = \text{SSB}/\text{SST}$, the proportion of variance explained by the group factor.

## Assumptions

- Independent observations within and across groups.
- Normal residuals (or large enough $n_i$).
- Homogeneous variances across groups (Welch ANOVA relaxes this).

## R Implementation

```r
library(effectsize); library(broom); library(car)
set.seed(2026)

df <- data.frame(
  group = factor(rep(c("A", "B", "C", "D"), each = 30)),
  y     = c(rnorm(30, 50, 8),
            rnorm(30, 55, 8),
            rnorm(30, 58, 8),
            rnorm(30, 54, 8))
)

# Omnibus ANOVA
```

```r
fit <- aov(y ~ group, data = df)
summary(fit)
tidy(fit)

# Assumption checks
leveneTest(y ~ group, data = df)
shapiro.test(residuals(fit))

# Effect size
eta_squared(fit)
omega_squared(fit)

# Welch ANOVA (unequal variances)
oneway.test(y ~ group, data = df)

# Tukey HSD post-hoc
TukeyHSD(fit)
```

## Output & Results

```
             Df Sum Sq Mean Sq F value  Pr(>F)
group         3    891   297.0    4.83 0.00317 **
Residuals   116   7135    61.5

Eta2 (partial) | 95% CI
--------------------------
0.11           | [0.02, 1.00]

        Tukey multiple comparisons of means 95% family-wise confidence level

         diff     lwr    upr  p adj
B-A   5.421   -0.58  11.42 0.097
C-A   8.215    2.21  14.22 0.003
D-A   4.129   -1.87  10.13 0.287
C-B   2.794   -3.21   8.79 0.627
D-B  -1.292   -7.29   4.71 0.943
D-C  -4.086  -10.09   1.92 0.297
```

$F(3, 116) = 4.83$, $p = 0.003$, rejects equal means. Tukey identifies C vs. A as the significant pair.

## Interpretation

Report: "A one-way ANOVA showed a significant group effect ($F(3, 116) = 4.83$, $p = 0.003$, $\eta^2 = 0.11$). Tukey HSD comparisons indicated that group C differed significantly from group A (mean difference 8.2, 95 % CI 2.2 to 14.2, adjusted $p = 0.003$)."

## Practical Tips

- Always check homogeneity of variances; if violated, use Welch ANOVA.
- Always report an effect size; F alone is not enough.
- For many post-hoc comparisons, Tukey is the standard; Dunnett is for comparing treatments to a control; Bonferroni or Holm for pre-specified families.
- Non-parametric alternative: Kruskal-Wallis.
- For within-subjects data, use repeated-measures ANOVA or a linear mixed model.

## For Reviewers

**What to look for in a paper using this method.**

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

## See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests